

EWDC: Integrating DECT-2020 With Wi-Fi for Enhanced Wireless Direct Connectivity

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Abstract—The rapid growth of smartphones, laptops, and Internet of Things (IoT) devices has created an increasing demand for seamless, secure, and energy-efficient wireless direct communication. Current popular solutions, such as AirDrop and Intel Unison, rely on bluetooth low energy (BLE) for device discovery and pairing, alongside IEEE 802.11 (Wi-Fi) for high-speed data transmission. However, research has identified significant vulnerabilities in BLE, including information leakage and profile manipulation. To address these challenges, we propose an enhanced wireless direct connectivity (EWDC) framework that integrates digital enhanced cordless telecommunications (DECT)-2020 new radio (NR) with Wi-Fi. In this framework, DECT replaces BLE to enable secure and efficient device discovery, while Wi-Fi supports high-bandwidth data transmission. Specifically, DECT utilizes its licensed spectrum to establish a secure communication channel for sharing Wi-Fi configuration information, enabling a secure and reliable setup of high-speed Wi-Fi data transmission links between devices. To validate this approach, we implemented the DECT medium access control layer on the Nordic nRF9161-DK platform and evaluated the end-to-end delay between two devices. Our results demonstrate an impressive transmission delay of less than 1 μ s for EWDC communication, enabling rapid device discovery and highly efficient operation. Furthermore, to enhance the security of wireless direct connections, the WireGuard, a VPN tunnel, is employed to encrypt data transmission over Wi-Fi providing robust protection for data shared on public frequency bandwidths. The proposed hybrid DECT-2020 NR and Wi-Fi solution significantly improves reliability, security, and communication efficiency, positioning it as a robust candidate for next-generation IoT communication systems.

Index Terms—DECT-2020, digital enhanced cordless telecommunications (DECT), enhanced wireless direct connectivity (EWDC), wireless direct communication.

I. INTRODUCTION

WITH the rapid development of Internet of Things (IoT) technologies and applications, the demand for wireless direct communication has grown significantly to enable seamless information exchange between devices over noncellular networks. To meet this need, wireless direct communication in the IoT context must provide both efficient device discovery and high-speed data transfer. Prominent examples include

apple wireless direct link (AWDL),¹ which supports over two billion devices across multiple operating systems via AirDrop, and Intel Unison, which enables direct file transfers, calls, messaging, and notifications between Windows PCs and Android/iOS devices.² These solutions typically rely on bluetooth low energy (BLE) for device discovery and pairing, followed by IEEE 802.11 (Wi-Fi) for high-speed data transmission.

While vendor-specific solutions like AWDL and Intel Unison provide convenience and cohesive user experiences, they fail to address critical challenges related to security, privacy, and interoperability in the expanding IoT landscape [1], [2], [3], [4], [5], [6], [7]. For instance, AWDL is vulnerable to man-in-the-middle (MitM) attacks, allowing attackers to intercept sensitive information, such as phone numbers and email addresses, during Apple AirDrop transmissions [2]. Although methods, such as private set intersection, proposed in [3], enhance mutual authentication with subsecond delays, they are limited to vendor-specific ecosystems and do not extend to broader IoT applications that require secure and efficient device discovery. Similarly, Intel Unison, despite its broader device and platform support, relies on BLE-based advertisements for device discovery, which operate on public frequencies and open protocols, exposing them to vulnerabilities like MitM attacks, session hijacking, and data leakage [4], [5]. Persistent weaknesses in BLE, including those identified in [6], span its physical, firmware, and host layers, with 76 documented attacks and 33 countermeasures highlighting its limitations. Nevertheless, solutions, such as Bluetooth 5.4's advertising encryption and the lightweight security frameworks proposed in [7] aim to enhance device discovery and reduce interception risks, but they have yet to be standardized or widely implemented in practical IoT environments. Therefore, there is a need for alternative technologies that combine communication efficiency, security, and scalability for wireless direct connectivity in IoT scenarios.

The digital enhanced cordless telecommunications (DECT)-2020 new radio (NR), developed by the European Telecommunications Standards Institute (ETSI), is specifically designed to meet the increasing demands of massive machine-type communication (mMTC) and ultrareliable low-latency communication (URLLC), which are essential for Industry 4.0 and IoT applications.³ Recognized by the ITU as part

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¹<https://support.apple.com/en-us/119857>

²<https://www.intel.com/content/.../unison/overview.html>

³<https://www.etsi.org/technologies/dect>

of the IMT-2020 technology family, DECT-2020 operates in both licensed and unlicensed spectrum below 6 GHz, providing flexibility for diverse deployment scenarios. It also incorporates support for ultra low energy (ULE) in its future evolution [8]. With over 20 years of development since its origins alongside the global system for mobile communications (GSM), DECT has evolved into a robust standard offering advanced security features like SIM authentication, making it highly secure for IoT environments. The combination of licensed spectrum options, ULE support, and strong security mechanisms establishes DECT-2020 as a compelling choice for scalable and reliable wireless direct communication in demanding IoT scenarios.

Despite the advancements proposed by the DECT-2020 standard, its development on chipsets has not fully kept pace with its specified capabilities. For instance, the DECT-2020 standard defines a new physical interface based on cyclic prefix orthogonal frequency-division multiplexing (CP-OFDM), capable of achieving data rates up to 1.2 Gb/s with 1024-QAM modulation [9]. However, the actual data rates achievable with chipsets like the nRF9161 depend on various factors, including specific implementations, configurations, and application requirements, and often fall short of the theoretical maximum. In comparison, Wi-Fi standards like IEEE 802.11be (Wi-Fi 7) achieve significantly higher data rates of up to 40 Gb/s [10] and are supported by well-designed chipsets from major firms, such as Qualcomm.⁴ Therefore, integrating DECT-2020 with Wi-Fi offers the potential to leverage the strengths of both technologies, enabling more robust and efficient wireless direct connectivity solutions for large-scale IoT scenarios.

To this end, we propose an enhanced wireless direct connectivity (EWDC) framework that integrates DECT-2020 with Wi-Fi, replacing BLE with DECT-2020 for secure and efficient device discovery, while Wi-Fi continues to support high-bandwidth data transmission. By operating IoT devices on DECT's licensed spectrum, the framework avoids interference in public channels, enabling faster and secure device discovery. Furthermore, the DECT standard cipher enhances device authentication and encrypts wireless connections between DECT devices, ensuring the protection of sensitive Wi-Fi configuration information. To address potential data leakage during Wi-Fi transmission on public frequencies, the framework incorporates WireGuard, which provides a secure Layer 3 VPN tunnel for fast and secure data sharing [11]. Together, these DECT-2020 design and integrations deliver a more secure and robust wireless direct connectivity solution for IoT scenarios. The main contributions of this article are summarized as follows.

- 1) First, we propose the EWDC framework, which integrates DECT-2020 with Wi-Fi in a manner similar to AWDL. However, unlike AWDL, DECT-2020 replaces BLE to provide secure and efficient device discovery, while Wi-Fi handles high-bandwidth data transmission. The framework utilizes DECT's licensed spectrum to reduce interference in public channels, ensuring robust

device authentication and encrypted communication. This design enables Wi-Fi to focus on transmitting large volumes of data efficiently.

- 2) Second, the DECT-2020 medium access control (MAC) layer is implemented on the Nordic nRF9161-DK platform to enable practical validation and real-world application of the framework. Specifically, the developed MAC layer adheres to DECT-2020 standards and supports beacon, association, and user data messages. These messages facilitate device discovery, pairing, and synchronization, operating on DECT-specific licensed frequencies to minimize interference and ensure reliable communication.
- 3) Third, a DECT-2020 sniffer and corresponding Wireshark dissectors are developed to analyze DECT-2020-specific packets, thereby facilitating MAC protocol verification and debugging. These tools enhance both development efficiency and reliability, and they are publicly available as open-source on GitHub, along with introductory documentation and DECT-2020 packet captures.⁵
- 4) Fourth, leveraging the developed MAC layer and Wireshark dissectors, the EWDC framework is evaluated through a real-world implementation by integrating DECT-2020 with Wi-Fi. Because Wi-Fi operates over public channels and is susceptible to MitM attacks, WireGuard is incorporated to protect the EWDC system. This integration enables automatic configuration and establishes a VPN tunnel, thereby preventing data leakage during Wi-Fi data transmissions.

The remainder of this article is organized as follows. Section II reviews the DECT-2020 standard, highlighting its new features in the PHY and MAC layers, and provides a comparative study of DECT-2020, BLE, and Wi-Fi. Section III introduces the EWDC framework, detailing its design and the integration of DECT-2020 with Wi-Fi. Section IV demonstrates the implementation of DECT-2020, including the developed MAC layer, DECT-2020 sniffer, and Wireshark dissectors. Section V evaluates the performance of the EWDC framework in terms of latency, jitter, and bandwidth, as well as the impact of WireGuard on ping and data transmission speed. Finally, Section VI concludes this article and outlines directions for future work.

II. BACKGROUND ON DECT-2020 STANDARDS

This section examines the DECT-2020 PHY and MAC layers and presents a comparative analysis of DECT-2020 against BLE and Wi-Fi.

A. DECT-2020 Specifications

The DECT-2020 NR specifications comprise a series of key technical standards that collectively define the system's capabilities and architecture. An overview of DECT-2020 NR is proposed by the ETSI in [12], introducing fundamental concepts and system features. The details of radio reception and

⁴<https://www.qualcomm.com/products/technology/wi-fi/wi-fi-7>

⁵<https://github.com/Aalto5G/DECT-NR-Wireshark-dissector>

transmission are specified in [13], while the physical (PHY) layer specifications are elaborated in [14], providing the technical requirements for signal processing and modulation. The document in [15] outlines the medium access control (MAC) layer requirements, defining how devices access the shared communication medium. Finally, the data link control (DLC) and convergence layers are described [16], which manage data transmission reliability and integration with higher-layer protocols. Unlike traditional 3GPP 5G standards, DECT-2020 introduces dynamic device roles, enabling devices to operate as fixed termination points (FT), Portable Termination points (PT), or in dual roles, simplifying network setup and allowing decentralized, self-healing networks. This combination of adaptability, scalability, and robust performance positions DECT-2020 as a secure and efficient communication solution for the 5G era [9], [17].

B. PHY Layer

The DECT-2020 PHY layer employs advanced technologies, such as CP-OFDM, time division duplex (TDD), turbo coding, hybrid automatic repeat request (HARQ), and multiple-input-multiple-output (MIMO) with up to eight streams. Operating primarily in the 1880–1900 MHz range in Europe and equivalent license-exempt bands globally, it supports subcarrier spacing from 27 kHz to 221.184 kHz, a link budget of 124 dB, and adjustable transmit power from –40 to 23 dBm, ensuring robust and low-latency communication. With bandwidth options of 1.728 MHz, 3.456 MHz, and 6.912 MHz, the PHY layer accommodates 12 modulation and coding schemes (MCS), ranging from the highly robust BPSK (code rate 1/2) to the high-capacity 1024-QAM (code rate 5/6). It also offers flexible numerologies for adjusting subcarrier spacing and Fourier transform scaling. PHY layer packets are structured into four parts: 1) synchronization training field (STF); 2) demodulation reference signal (DRS); 3) control channel; and 4) data channel. Each frame spans 10 ms and is divided into 24 slots, which can be further subdivided into subslots based on the selected numerology.

C. MAC Layer

The DECT-2020 MAC layer supports both deterministic scheduled transmissions and random access, providing flexibility to handle diverse traffic types. Scheduled transmissions are managed via the dedicated channel (DCH), where FT-mode radio devices (RDs) allocate radio resources for PT-mode RDs requiring predictable latency, such as URLLC traffic. For unscheduled and asynchronous traffic, the random access channel (RACH) employs listen-before-talk (LBT) with a minimum sensing time of two symbols, enabling rapid medium access while minimizing collisions. Additionally, RSSI-based interference assessments further ensure coexistence with legacy systems, maintaining deterministic connectivity [18]. Furthermore, both deterministic and random access traffic can utilize a common PHY layer packet format with HARQ for reliable transmission, while alternative formats cater to specific service needs. In addition to DCH and RACH, DECT-2020 MAC layer also provide the broadcast channel (BCH) and

paging channel (PCH) for control signaling and device discovery, facilitating reliable point-to-point and point-to-multipoint wireless links. In point-to-multipoint scenarios, FT-mode RDs specify slots and subslots for RACH access, assign MCS, and manage packet durations to optimize resource utilization. To facilitate DECT-2020 applications, the MAC layer provides a dedicated traffic channel (DTCH) and a multicast traffic channel (MTCH) for upper layer, enabling efficient resource allocation and making it ideal for mMTC requiring low latency and high reliability.

D. DECT-2020 Standards Evaluation

Several studies have examined the DECT-2020 standard to assess the performance and applicability of its PHY and MAC layers for mMTC.

The work-in-progress evaluation of DECT-2020 presented in [19] tested its performance at 450, 1350, and 3700 MHz with bandwidths ranging from 1.728 to 13.824 MHz. packet error rates (PERs) were reported as less than 5% at 450 MHz, 15% at 1350 MHz, and 10% at 3700 MHz, with higher errors attributed to hardware losses. Additionally, in [20], a measurement-based study assessed DECT-2020 link distances in indoor and outdoor scenarios using precommercial equipment, and the results showed indoor LOS ranges of 40–190 m and NLOS distances over 60 m with TX power up to 0 dBm, while outdoor LOS distances reached up to 600 m at 19 dBm. Further insights are provided in [21], where the DECT-2020 NR PHY layer was studied in industrial environments under various channel conditions; despite its promising design, simulations revealed that production line communications did not meet the required PER threshold, and security camera applications also exhibited higher error rates than expected. Collectively, these findings underscore the need for a robust MAC layer integrating link control techniques and enhanced access mechanisms to ensure reliable performance in industrial applications.

For MAC-layer enhancements, [22] introduced an NS-3⁶ simulator featuring *DectMac* and *DectPhy* layers, enabling RD node creation, association, and evaluated packet outage rates in mMTC scenarios. Performance requirements for URLLC and mMTC were further investigated in [23], while [24] proposed positioning methods leveraging DECT-2020 with time difference of arrival, and [25] explored a direction of arrival-based indoor localization approach. Energy efficiency considerations in wide-range battery-operated mesh networks were highlighted in [26] and [27], underscoring the demand for DECT-ULE. To enhance MAC performance, [28] presented a protocol aiming to reduce outage probability below 1% by employing beacon messages for efficient data routing within a region of interest. Additionally, [29] investigated relay and next-hop selection algorithms to boost packet delivery success in DECT-2020 multihop networks. Despite the promising results from simulation studies demonstrating the advanced features of DECT-2020, they lack real-world implementations, where dynamic and variable conditions pose greater challenges than controlled environments. To address this gap, this article

⁶<https://www.nsnam.org/>

TABLE I
COMPARISON OF DECT WITH BLE AND WI-FI

Parameter	BLE	Wi-Fi 6	DECT-2020
Standard	Bluetooth SIG	IEEE 802.11ax	ETSI DECT-2020
Frequency	2.4 GHz	2.4/5/6 GHz	1.88–1.9/2.4/5/6 GHz
PHY Layer	FHSS	OFDMA	CP-OFDM
MAC Layer	Polling	CSMA/CA	TDMA
Speed	<2 Mbps	<9.6 Gbps	<1.2 Gbps
Power (dBm)	–20 ~ 20	–20 ~ 30	–20 ~ 23
Encryption	AES-128	WPA3	DECT Standard Cipher
Mesh Support	Yes	Optional	Yes
Active Users	7	256	24
Price	Low	Moderate	Low-Moderate

presents the development of the DECT-2020 MAC layer and a DECT-2020 sniffer to validate its performance in practical scenarios.

E. Comparative Study of DECT-2020, BLE, and Wi-Fi.

Table I provides a comparison of DECT-2020 with BLE and Wi-Fi 6, emphasizing their characteristics and origins. BLE, originally developed by Nokia, is widely recognized for its energy efficiency and cost-effectiveness. Standardized by the Bluetooth special interest group (SIG), BLE operates in the 2.4 GHz ISM band, employing frequency hopping spread spectrum (FHSS) at the PHY layer and polling at the MAC layer, making it suitable for low-throughput and direct connections [30]. Wi-Fi, defined under the IEEE 802.11 standards, operates across the 2.4/5/6 GHz ISM bands. Wi-Fi 6 employs orthogonal frequency-division multiple access (OFDMA) at the PHY layer to achieve high-bandwidth data rates of up to 9.6 Gb/s [31]. However, its use of CSMA/CA at the MAC layer introduces contention and random delays, limiting its deterministic performance.

In contrast, DECT-2020 NR can operate in both the licensed band and unlicensed ISM bands below 6 GHz. It uses CP-OFDM at the PHY layer and time division multiple access (TDMA) at the MAC layer, achieving a maximum data rate of 1.2 Gb/s while ensuring deterministic performance. DECT-2020 also offers enhanced mesh support through dual FT and PT modes, making it professional for industrial IoT scenarios. For security, DECT-2020 relies on its Standard Cipher, while BLE employs AES-128 and Wi-Fi 6 implements WPA3. In terms of point-to-multipoint communication, BLE supports up to seven active connections, Wi-Fi 6 scales to 256 users, and DECT-2020 accommodates up to 24 active slaves through its time-slot-based allocation. Overall, DECT-2020's deterministic communication, low latency, and robust mesh networking enabled by its dual FT and PT modes make it superior to BLE and Wi-Fi for time-critical applications like industrial automation and robotics in IoT scenarios.

III. ENHANCED WIRELESS DIRECT CONNECTIVITY

This section introduces the EWDC framework, which leverages DECT-2020 for device discovery and Wi-Fi settings exchange, followed by secure data transfer over Wi-Fi with VPN encryption.

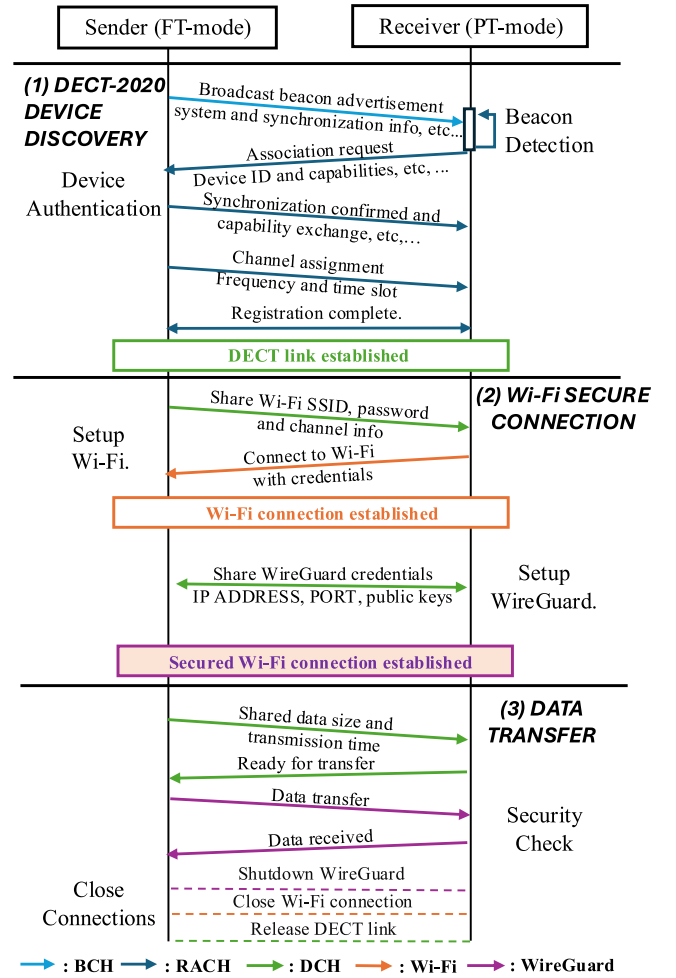


Fig. 1. Proactive EWDC Data Transmission.

A. DECT-2020 Device Discovery

Fig. 1 illustrates the proactive EWDC data transmission framework, in which DECT-2020 replaces BLE to enhance synchronization, extend communication range, and improve device discovery reliability. In this configuration, the sender operates in the FT-mode, periodically broadcasting beacons containing target device information, while the receiver functions in the PT-mode, scanning for these beacons to establish a secure and reliable DECT link over a licensed frequency and channel.

During the discovery phase of the proactive EWDC data transmission, the FT-mode sender broadcasts beacons on the BCH. These beacons carry network parameters, including network IDs, available frequencies, time-slot configurations, and security details. The PT-mode receiver scans the beacons, identifies a suitable network, and verifies whether the beacon indicates its own device ID as the intended target. If the receiver opts to connect, it transmits an association request to the FT node, which validates the request and allocates initial resources. Upon confirming the association, the PT node synchronizes its internal clock with the FT node's timing signals embedded in the beacon transmissions. This synchronization relies on the STF, ensuring data transmissions

occur within designated time slots, thereby minimizing collisions and maximizing spectral efficiency. Subsequently, the PT node provides its device information and capabilities for final verification by the FT node. After successful validation, the FT node dynamically assigns channels and time slots based on network conditions, interference levels, and device capabilities. During registration, the FT node refines these assignments by dividing each TDMA frame into intervals for messaging, acknowledgments, and security procedures, such as key exchange. This association leverages the RACH mechanism in DECT-2020 to reduce interference and ensure reliable connectivity.

In the reactive EWDC data transmission mode, the sender operates in PT mode while the receiver functions in FT mode. During the DECT discovery phase, the PT-mode sender detects beacons broadcast by the FT-mode receiver, verifies that its target device ID is included, and directly sends an association request. The subsequent steps, including association, synchronization, channel assignment, and registration, proceed similarly to the proactive mode. By employing a systematic process of beacon broadcasting, association, synchronization, channel assignment, and dynamic frequency and time-slot allocation, DECT-2020 ensures reliable and deterministic communication. These well-defined procedures enable efficient spectrum utilization, low-latency performance, and deterministic operation, making DECT-2020 ideally suited for industrial environments requiring robust connectivity.

B. Wi-Fi Secure Connection

Once the DECT link is established, EWDC initiates a secure Wi-Fi connection. During the Wi-Fi setup phase, the sender shares network credentials, including the SSID, password, and channel information, over the DECT DCH and instructs the receiver to connect. At the same time, the sender sets up or activates a Wi-Fi hotspot using these credentials. On the receiver side, the shared Wi-Fi information is used to either join the hotspot or establish a direct Wi-Fi link. Alternatively, a Wi-Fi P2P network can be formed using Wi-Fi Direct, requiring both the sender and receiver to exchange key connection details, such as the device name, capabilities, network SSID, security credentials (such as a passphrase or WPS configuration), role designation (Group Owner for the sender or Client for the receiver), and channel information.

Wi-Fi operates on public frequency bands, making it vulnerable to attacks; to enhance security, EWDC integrates WireGuard after establishing the Wi-Fi connection. Each device (sender and receiver) generates a private and public key pair, securely exchanging their public keys along with IP addresses and port information over the DECT DCH. With features like scheduling, acknowledgments, and retransmission requests, DECT DCH ensures that the key exchange is accurate and remains protected from exposure on the public network. Using the exchanged public keys and connection details, both devices update their WireGuard configuration files to include the other as a peer, enabling the creation of an encrypted tunnel. This approach combines WireGuard's

encryption with DECT DCH's secure transmission, protecting data over public Wi-Fi from vulnerabilities.

C. Data Transfer

During data transfer, EWDC combines the DECT DCH and a WireGuard-secured Wi-Fi link for robust and efficient communication. The DECT DCH serves as a reliable control channel, allowing the sender to share metadata, such as file size and estimated transmission time, before the data transfer begins. This preparatory step ensures the receiver can allocate resources effectively, supporting a more deterministic and efficient process. The actual data is transmitted through the encrypted WireGuard tunnel, where each device's IP address is authenticated using previously exchanged public keys while unverified packets are rejected to prevent unauthorized access and malicious data injection. By using the DECT DCH as a control channel for transmission scheduling and status updates, the EWDC minimizes exposure of the Wi-Fi channel on public bands, significantly reducing the risk of eavesdropping or sniffing attempts. Once the data transfer concludes or exceeds the predefined time limit, the system immediately shuts down the WireGuard tunnel, minimizing opportunities for attackers to analyze or decrypt intercepted packets. Following this, all Wi-Fi credentials are invalidated, any remaining information unusable to potential adversaries. Finally, the DECT link is terminated to free resources for other devices and improve spectral efficiency.

Overall, the proposed EWDC framework combines DECT-2020 for efficient device discovery with a WireGuard-secured Wi-Fi link for secure data transfer. DECT-2020 establishes reliable connections through beaconing, synchronization, and channel assignment, minimizing interference and ensuring predictable performance. The WireGuard-secured Wi-Fi link leverages the DECT DCH for exchanging credentials and keys, protecting data traffic via the encrypted tunnel. This integration ensures data confidentiality and integrity while minimizing security vulnerabilities, making it a robust solution for industrial and IoT scenarios that require secure and reliable communication.

IV. EWDC IMPLEMENTATION

To support testing, we implemented the basic DECT-2020 MAC layer functionality on Nordic development tools, enabling the setup and evaluation of the proposed EWDC framework. Additionally, a DECT-2020 packet sniffer and custom Wireshark dissectors were developed to analyze DECT-2020-specific packets, facilitating protocol verification and debugging. A WireGuard-secured Wi-Fi link was implemented to ensure secure data transmission within the EWDC system. The overall efforts are summarized as follows.

A. DECT-2020 Implementation

The DECT nRF9161 development kit, provided by Nordic Semiconductor,⁷ was used to verify the proposed EWDC system. Specifically, its 1.9 GHz band support and optimized

⁷<https://www.nordicsemi.com>

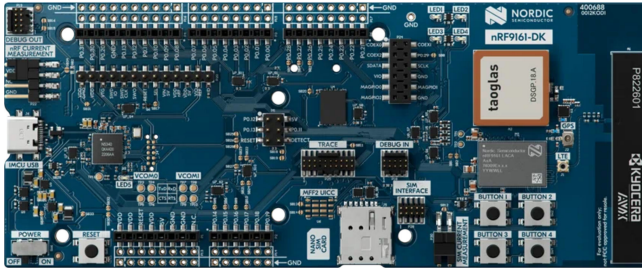


Fig. 2. DECT-2020 nRF9161 development kit.

modem features make it a scalable and efficient solution for DECT-2020 evaluation. Additionally, the nRF9161 SiP integrates LTE-M/NB-IoT and GNSS, enhancing its suitability for commercial deployment with global IoT connectivity. As shown in Fig. 2, the kit integrates the Zephyr real-time operating system (RTOS), which interfaces with the board's chipsets, drivers, and libraries to support a variety of embedded applications and the DECT-2020 radio protocol stack. The kit also includes MCUBoot, a secure bootloader that ensures a trusted environment during DECT communication setup, further enhanced by middleware, libraries, and hardware drivers. By leveraging the nRF toolchain and nRF Util tools,⁸ we can easily interact with the chipsets, enabling robust and efficient implementation of the DECT-2020 FT and PT MAC layer.

1) *FT MAC Layer Implementation*: The FT MAC layer manages radio resources, handles association procedures, and coordinates data exchange with PTs. It begins with channel scanning to identify an optimal cluster channel based on RSSI metrics, ensuring reliable communication with minimal interference. The FT then periodically transmits cluster beacons to announce its presence and provide RACH parameters for potential PT associations. Upon receiving an association request, the FT allocates dedicated subslots, constructs an association response with resource allocation IEs, and schedules subsequent transmissions to support user data and keep-alive exchanges, maintaining active connections effectively.

Fig. 3 illustrates the FT MAC layer operating through a state machine that systematically organizes and sequences its MAC functionalities. The process starts in the *Beaconing* state, where the FT transmits cluster beacons to advertise RACH parameters and establish synchronization. After completing the beacon transmission, the FT transitions to the *RachRX* state, where it monitors the shared channel for association requests from PTs or checks for existing data resources. If a valid association request is detected, the state machine moves to the *RachRes* state, where the FT constructs and transmits an association response, including resource allocation. The FT processes multiple association requests in a loop, returning to *RachRX* or *Beaconing* if no additional requests are pending or all responses are sent. If no association is required, the FT directly transitions to the *DataRX* state to handle uplink data. Once in the *DataRX* state, the FT receives uplink data from associated PTs, transitioning to

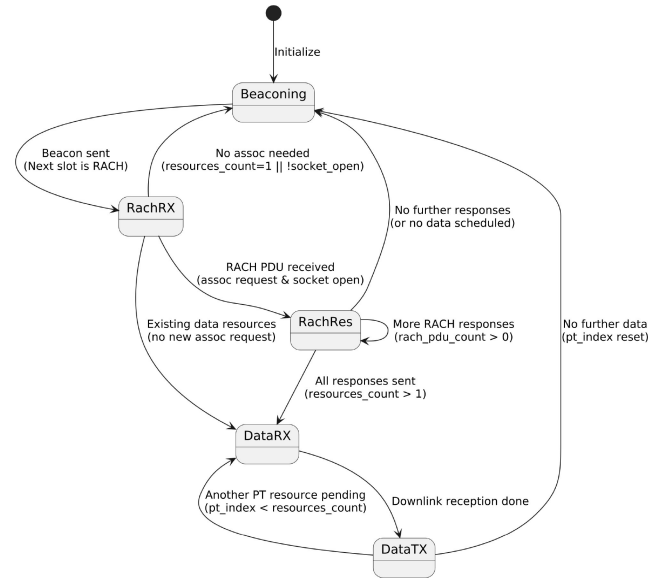


Fig. 3. FT MAC layer state machine.

the *DataTX* state to transmit downlink data or keep-alive messages. After completing data transfers, the FT evaluates whether additional PT resources are pending, transitioning back to *DataRX* if required or returning to the *Beaconing* state to restart the cycle. This modular state machine efficiently manages beaconing, association handling, and data transfers, ensuring reliable and structured operation of all DECT MAC functionalities on the FT side. The transitions between states are dynamically determined by resource availability, pending events, and operational conditions, maintaining robust and synchronized communication.

In addition to the state machine, the FT MAC layer introduces a flexible memory pool architecture that supports dynamic storage, queuing, and processing of both inbound and outbound frames without relying on hardware-specific constraints. For example, a resource table is implemented on the FT side, using the short RD IDs of each PT as keys to map specific communication resources, such as allocated time slots, repetition patterns, and channel assignments. Additionally, the FT maintains an association table to track essential information for each PT, including status, expiration times, and allocated flows. This architecture enables the FT to efficiently manage multiple associations, reallocate resources when necessary, and release them as needed. By leveraging the memory pool, resource table, and association table, the FT achieves fine-grained control over the scheduling of uplink and downlink transmissions, ensuring seamless and efficient operation of PTs.

2) *PT MAC Layer Implementation*: The PT MAC layer is responsible for discovering and associating with an available FT and exchanging data through scheduled subslot allocations provided by the FT. Initially, the PT listens for periodic cluster beacons from the FT to acquire channel parameters and determine the timing and method for initiating random access. Upon receiving a target beacon, the PT sends an association request. If the request is acknowledged, the FT

⁸<https://www.nordicsemi.com/Products/Development-tools/>

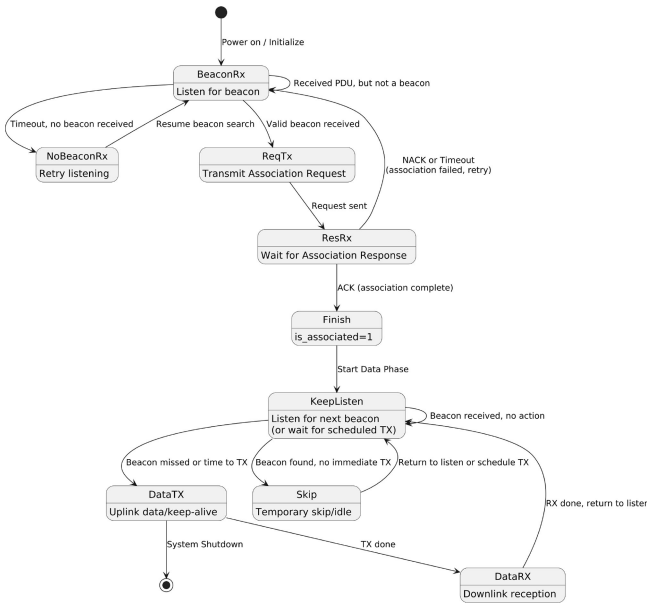


Fig. 4. PT MAC layer state machine.

allocates dedicated resources, enabling the PT to perform seamless uplink and downlink communication.

Fig. 4 illustrates the PT MAC layer operating through a structured state machine that systematically manages beacon discovery, association requests, and data exchange with the FT. During beacon discovery, the PT starts in the *BeaconRx* state, which can transition to one of three outcomes: 1) the *NoBeaconRx* state if no beacon is detected within the timeout period; 2) a loop back to the *BeaconRx* state to continue searching for a valid beacon; or 3) progression to the *ReqTx* state to initiate an association request upon receiving a valid beacon. Once the association starts, the the *ReqTx* state transmit an association request to the FT and then transitions to the *ResRx* state to await an association response. If the response is acknowledged (ACK), the PT receive enters the *Finish* state. In case of a negative ACK (NACK) or timeout, the PT returns to the *BeaconRx* state to retry the association process. After association, the PT transitions to the *KeepListen* state, where it listens for the next beacon or waits for scheduled transmission slots from the FT. Upon receiving an uplink opportunity, the PT moves to the *DataTX* state to transmit data or keep-alive messages, followed by transitioning to the *DataRX* state to receive downlink data from the FT. Alternatively, the PT may enter the *Skip* state to idle for energy saving, temporarily ignoring the transmission schedule from the FT. Once data reception or the idle period is complete, the PT returns to the *KeepListen* state, repeating the communication cycle. Throughout these states, the PT aligning its subslot timing with the FT's beacon reference, ensuring that both sides operate on a shared timeline, thereby reducing the risk of collisions and synchronization errors.

In addition to the state machine, the PT MAC layer employs a streamlined buffering system and simple flags to efficiently manage uplink/downlink data and keep-alive traffic with its associated FT. Specifically, the PT uses a global buffer for user-plane data intended for socket transmissions and a

dedicated static buffer for keep-alive messages. This separation eliminates contention between regular uplink packets and critical keep-alive frames. Additionally, the PT maintains a designated receive buffer to store incoming data payloads from the FT, preserving the buffer until higher layer processes retrieve and acknowledge the data. To facilitate traffic management, the PT employs simple flags, such as *is_data_sending* and *is_data_received*, to indicate the readiness to send user data or keep-alive protocol data units (PDUs). By leveraging these buffers and flags, the PT ensures efficient handling of diverse traffic types, sufficient buffering capacity, and timely packet delivery.

The DECT MAC layer, including both FT and PT, interfaces with the dedicated PHY layer via well-defined transmit/receive APIs provided by the nRF9161. Instead of embedding hardware-specific details within the MAC code, the MAC layer utilizes function calls, such as *dect_nr_phy_transmit()* and *dect_nr_phy_receive()*, to execute low-level transmission and reception operations. These APIs encapsulate the complexities of modulation, timing synchronization, and physical channel access, enabling the MAC layer to focus on higher-level responsibilities, such as scheduling beacons, allocating resources, and handling association requests. PHY callbacks, in turn, provide feedback to the MAC layer with critical events (e.g., “transmission complete,” “reception complete,” or “beacon detected”), allowing the state machine to update internal structures and plan subsequent actions. The API-driven design of the nRF9161 ensures a modular, platform-independent MAC layer, enabling easy adaptation and reuse across hardware platforms with minimal changes.

B. DECT-2020 Sniffer and Wireshark Dissectors

To facilitate protocol verification and debugging of DECT-2020 specific packets, a dedicated DECT-2020 sniffer was developed based on the nRF9161 PHY layer API. Additionally, a Wireshark dissector was created to further support and streamline DECT-2020 development and analysis.

The DECT-2020 sniffer utilizes the PHY layer API provided by the nRF9161 to monitor specific DECT-2020 channels and capture raw PHY layer packets for subsequent MAC layer analysis. On the nRF9161 board side, the process starts with initializing the PHY layer parameters, including the start time, scanning duration, and callback functions. Operating in an infinite loop, the sniffer captures PHY packets, processes them through the *on_packet_received* callback function to skip PHY headers, and outputs the raw payloads in hexadecimal format. After each packet is processed, the *on_rx_complete* callback function reschedules the next receive request, ensuring continuous monitoring. On the sniffer's computer side, a UDP server-client setup connects the nRF9161 board via USB serial. The client reads the hexadecimal packets, encapsulates them into UDP packets, and forwards them to the server, where Wireshark⁹ monitors the communication port for package analysis.

⁹<https://www.wireshark.org/>

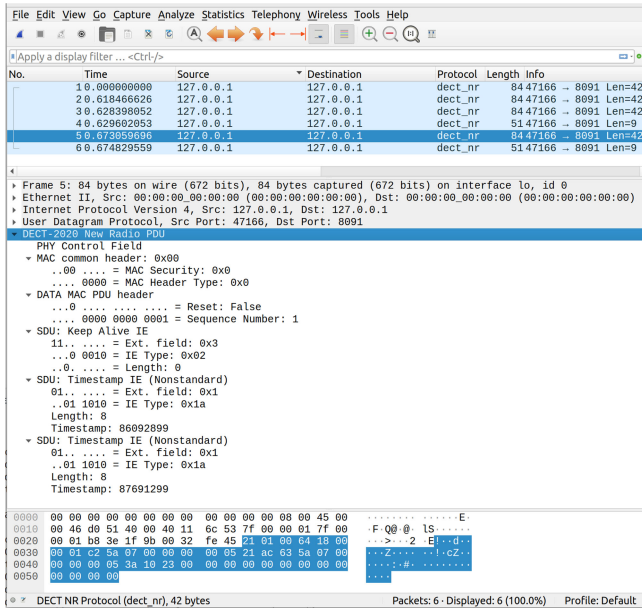


Fig. 5. Wireshark capture of Keep-Alive IE from PT.

Fig. 5 showcases a DECT-2020 NR PDU in Wireshark, revealing its layered structure. The PHY Control Field contains low-level physical parameters, while the MAC Common Header indicates that no security is applied and identifies this as a Data MAC PDU. The subsequent header layer shows a sequence number of 1 and no reset flag. Within the service data unit (SDU), a Keep-Alive IE is present to confirm that the link between the FT and PT remains active even in the absence of traffic. Additionally, two custom Timestamp IEs are included: one records the clock tick on the FT when the last packet was sent to the PT, which the PT sends back to the FT in the current packet, and the other records the clock tick on the PT when the current packet is sent to the FT. These custom Timestamp IEs are utilized in the subsequent testing phase measure DECT-2020 communication latency and jitter.

In addition to showcasing the Keep-Alive IE package, the developed sniffer and Wireshark dissector can capture the entire communication process between the FT and PT. This includes the initial beacon broadcasting on the BCH, association packets on the RACH, and various types of data transmitted on the DCH. These captured packets are recorded and published as a pcapng file on the Aalto 5G GitHub repository. Although the sniffer provides minimal decoding, its ability to dump raw payloads in hexadecimal allows seamless integration with tools like Wireshark. Additionally, its flexibility to monitor various DECT channels by modifying the carrier parameter makes it a powerful and versatile tool for DECT-2020 traffic analysis.

C. WireGuard-Secured Wi-Fi

In the EWDC framework, Wi-Fi is configured for data transmission after establishing a DECT-2020 connection. Since Wi-Fi operates on public spectrum and is susceptible to challenges, such as congestion and interference, DECT-2020 leverages its licensed spectrum to securely share Wi-Fi

```
[Interface]
# Client WireGuard interface
PrivateKey = <private-key-Client>
Address = 10.10.0.22/24

[Peer]
# Server peer configuration
PublicKey = <public-key-Server>
Endpoint = 10.42.0.1:51820
AllowedIPs = 0.0.0.0/0
PersistentKeepalive = 25
```

Listing 1. Client WireGuard Configuration.

```
[Interface]
# Server WireGuard interface
Address = 10.10.0.11/24
ListenPort = 51820
PrivateKey = <private-key-Server>

# NAT and forwarding rules for outbound internet
access via wlan0
PostUp = iptables -A FORWARD -i wg0 -j ACCEPT;
iptables -A FORWARD -o wg0 -j ACCEPT; iptables -
t nat -A POSTROUTING -o wlan0 -j MASQUERADE
PreDown = iptables -D FORWARD -i wg0 -j ACCEPT;
iptables -D FORWARD -o wg0 -j ACCEPT; iptables -
t nat -D POSTROUTING -o wlan0 -j MASQUERADE

[Peer]
# Client peer
PublicKey = <public-key-Client>
AllowedIPs = 10.10.0.22/32
```

Listing 2. Server WireGuard Configuration.

credentials, ensuring a reliable setup. To further secure data transmission over Wi-Fi, WireGuard is introduced to establish secure tunnels at Layer 3 of the network stack, thereby enhance the security data transmission.

WireGuard is used to establish a secure Layer 3 network tunnel, encrypting data traffic between peers, with the sender and receiver functioning as FT or PT. At its core, WireGuard associates a peer's public key with a tunnel source IP address. It uses a single round-trip key exchange based on the NoiseIK protocol and transparently manages session creation through a streamlined timer state machine [11]. Mutual authentication is achieved using short preshared static keys based on Curve25519 points, ensuring strong perfect forward secrecy and robust identity protection. For data encapsulation, WireGuard utilizes ChaCha20-Poly1305 authenticated encryption over UDP, delivering high transport speed and efficiency.

As shown in Listing 1 and Listing 2, the communication peers exchange public keys and tunnel source IP addresses via a dedicated DECT channel. These preshared keys and IP addresses are then used by WireGuard for mutual authentication and data encryption. Additionally, IP table configurations enable NAT and forwarding rules for outbound internet access over Wi-Fi. By incorporating these advanced security measures, EWDC ensures a secure, efficient, and reliable environment for wireless file transfers. This approach significantly enhances security compared to traditional BLE-based advertisement mechanisms, safeguarding user data throughout the entire communication process. The next section presents the EWDC testing results.

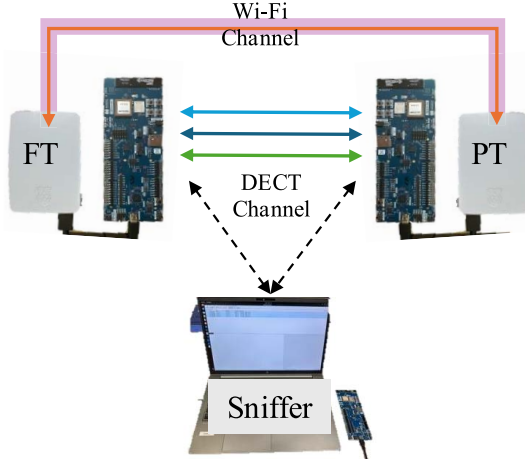


Fig. 6. EWDC setup.

V. EWDC TESTING

To evaluate the EWDC framework, a testing setup was designed using raspberry pi devices (RPis) and Nordic nRF9161 development kits. The testing focused on measuring DECT FT-PT delay and jitter, end-to-end delay and jitter on RPis, and the impact of WireGuard on ping latency and bandwidth in the EWDC system. Furthermore, a comparison between EWDC and the baseline AWDL method is presented, emphasizing aspects of latency, jitter, security, reliability, and energy efficiency.

A. Testing Setup

Fig. 6 presents the setup of the proposed EWDC system, which includes two RPis, three Nordic nRF9161 development kits, and a high-performance computer. Each RPi is connected to a DECT development kit operating in either FT or PT mode via USB serial. A sniffer is connected to one of the DECT development kits to capture packets transmitted over the DECT channel. Additionally, the RPis establish a Wi-Fi network and perform the WireGuard procedure, exchanging Wi-Fi credentials and WireGuard peers' keys and IPs over the DECT channel.

Table II provides an overview of the EWDC configuration parameters for the key devices in the system: DECT, RPi, and the computer. The DECT settings include the operating frequency of 1665 MHz, a channel bandwidth of 1.728 MHz, subcarrier spacing of 1728 kHz, a frame interval of 10 ms, and a power control level of 23 dBm, utilizing GFSK modulation. Communication with the DECT module is managed via UART with a baud rate of 115200 and a buffer size of 200 bytes. The RPi 5 serves as the central hardware platform for Wi-Fi communication and system integration, running the Debian operating system and utilizing the IEEE 802.11ac protocol on the 5.0 GHz band. The computer used in the configuration operates on Ubuntu 22.04, supporting system control and data processing. Together, these setup allowed for comprehensive testing of the EWDC performance and functionality in real-world scenarios.

TABLE II
EWDC CONFIGURATION PARAMETERS

Device	Parameter	Setting
DECT	Frequency Band	1665 MHz
DECT	Channel Bandwidth	1.728 MHz
DECT	Subcarrier Spacing	1728 kHz
DECT	Frame Interval	10 ms
DECT	Power Control	23 dBm
DECT	Modulation	GFSK
DECT	MODEM_TICK_RATE	69120000Hz
UART	Baud-rate	115200
UART	UART Buffer Size	200
RPi	Hardware	RPi 5
RPi	Operating System	Debian
RPi	Wi-Fi 802.11ac	5.0 GHz
Computer	Hardware	HP Zbook
Computer	Operating System	Ubuntu 22.04

B. Testing 1 (Connection Delay)

To measure the EWDC system's delay between the FT and PT, as well as the end-to-end delay between RPi-FT and RPi-PT, a beacon message is utilized. When the FT transmits a beacon (FT_send), it serves as the reference for the cluster beacon period and is embedded in the message sent to both the RPi connected to the FT ($RPi_PT_received$) and the PT ($PT_received_FT$). Upon receiving the beacon, the PT records its local time for the beacon reception and extracts the FT's beacon transmission time from the message. The PT then transmits a response back to the FT, including its recorded time (PT_send) and the FT's transmission time. The RPi connected to the PT also reads the buffer via the serial interface to retrieve the beacon message, recording its local time when the beacon is received ($RPi_PT_received$). Finally, the FT logs the timestamps for the responses it receives from both the PT ($FT_received$) and the RPi ($RPi_PT_received$), creating a comprehensive timeline for analyzing system delays. These time recording are raw presented in Table III.

In Table III, The DECT clock ticks are first converted into seconds using the modem's tick rate (69 120 000 Hz) with the formula:

$$\text{time in seconds} = \frac{\text{DECT ticks}}{\text{MODEM_TICK_RATE_HZ}}. \quad (1)$$

This conversion is applied to columns, such as FT_send , $PT_received_FT$, PT_send and $FT_received$, while for columns already in Unix time (e.g., $RPi_FT_received$ and $RPi_PT_received$) no further transformation is required. To align all timestamps on a common reference, the minimum timestamp across all columns is subtracted from each timestamp, effectively normalizing the data.

Fig. 7 illustrates the beacon transmission time difference between the FT and PT, measured in microseconds (μs). Specifically, the beacon transmission interval is set to 50 ms for this testing. The plot shows that the beacon transmission delay ranges between 0.6 and 0.9 μs , exhibiting periodicity.

TABLE III
EWDC SYSTEM TIMING LOGS

FT_send	RPi_FT_received	PT_received_FT	RPi_PT_received	PT_send	FT_received
36354172086	1730259977.565704	36307119922	1730171602.450699	36308718322	36355770526
36357628086	1730259977.615702	36310575922	1730171602.500698	36312174322	36359226526
36361084086	1730259977.665703	36314031922	1730171602.550659	36315630322	36362682526
36364540086	1730259977.715706	36317487922	1730171602.600705	36319086322	36366138526
36367996086	1730259977.765709	36320943922	1730171602.650708	36322542322	36369594526
...	...	...	...	...	...

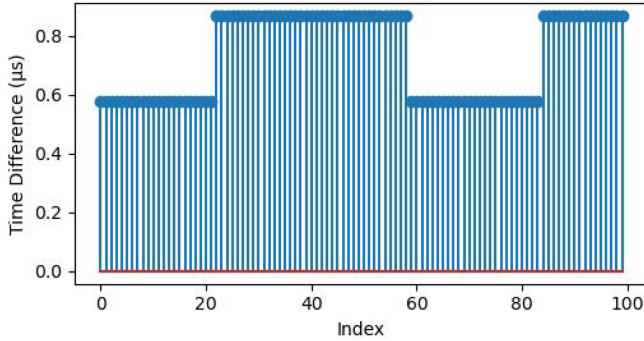


Fig. 7. Data transmission time between PT and FT.

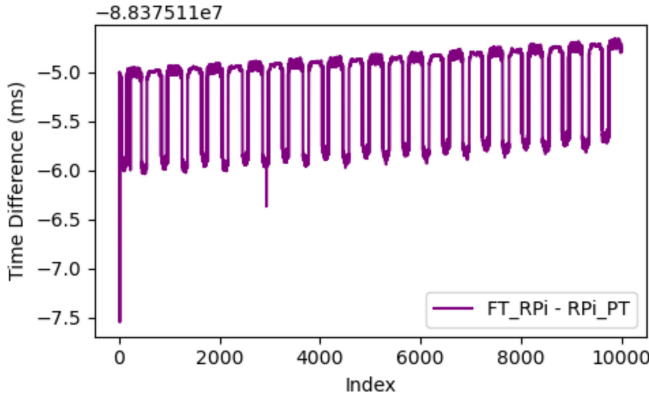


Fig. 8. Data transmission time difference between RPi.

The base delay of $0.6 \mu\text{s}$ arises from information processing at both the PT and FT, combined with signal transmission over the air. Although the PT synchronizes with the FT using the STF, the PT's onboard clock experiences drift, causing cumulative clock errors that require periodic adjustments to align with the FT's clock and maintain synchronization. Additionally, the Nordic nRF9161 development kit contains two separate chips: one for the DECT radio front-end and another as an embedded MCU running the Zephyr RTOS to process radio and MAC layer protocol stacks. These independent systems on the same board can introduce timing errors, impacting overall system performance.

Fig. 8 illustrates the data transmission time difference between RPi, measured in milliseconds (ms). The RPi is connected to the DECT boards via USB serial interfaces. The plot reveals that the time difference between the two RPi accumulates over time, with periodic fluctuations of approximately 1 ms. This behavior is primarily attributed to

the non real-time Linux operating system running on the RPi, which dynamically adjusts system interrupts to manage various system and user threads, leading to fluctuations and occasional large errors. Additionally, the RPi's hardware clocks exhibit lower precision, causing time discrepancies to accumulate over time. As all timestamps in the dataset are aligned to a common reference based on the minimum value across all columns, the y-axis reflects the biases in the RPi's time recordings.

Compared to RPi, DECT boards demonstrate superior timing precision, with transmission delays below $1 \mu\text{s}$. These delays exhibit minor periodic adjustments due to clock drift and synchronization at the Nordic nRF9161 development kit and the RTOS. In contrast, RPi shows larger, accumulating time delays with periodic fluctuations around 1 ms, primarily caused by the non real-time Linux operating system and lower-precision hardware clocks. To further assess their impact on EWDC system performance, connection jitter was measured separately across all system components.

C. Testing 2 (Connection Jitter)

Fig. 9 shows the jitter measurements and the corresponding histogram at FT and PT. The results in Fig. 9(a) and (b) indicate that jitter at both FT and PT remains below $0.3 \mu\text{s}$. Additionally, the jitter at FT exhibits greater variability compared to PT, primarily due to the FT's need to schedule transmission times for the PT and dynamically invoke processor for computation tasks. The histogram in Fig. 9(c) and (d) further confirm that most jitter values are below $0.1 \mu\text{s}$, with occasional errors causing values to exceed $0.3 \mu\text{s}$. These results highlight the superior timing precision of the DECT system.

Fig. 10 presents the jitter measurements and corresponding histograms for RPi_FT and RPi_PT. As shown in Fig. 10(a), the jitter at RPi_FT is mostly below 1 ms, while Fig. 10(b) indicates that RPi_PT initially experiences jitter exceeding 2 ms before stabilizing to levels similar to RPi_FT, generally below 1 ms. Both RPi_FT and RPi_PT exhibit periodic variations in jitter, peaking around 1 ms and returning to approximately 0.2 ms. This periodic behavior is likely due to system interrupts or periodic checks performed by the RPi to manage connected devices, leading to temporary processing delays. The histograms further illustrate that the majority of jitter events for RPi are concentrated below 0.2 ms, with occasional outliers indicating infrequent errors. This behavior is primarily due to the non real-time nature of the Linux OS on the RPi, which dynamically manages

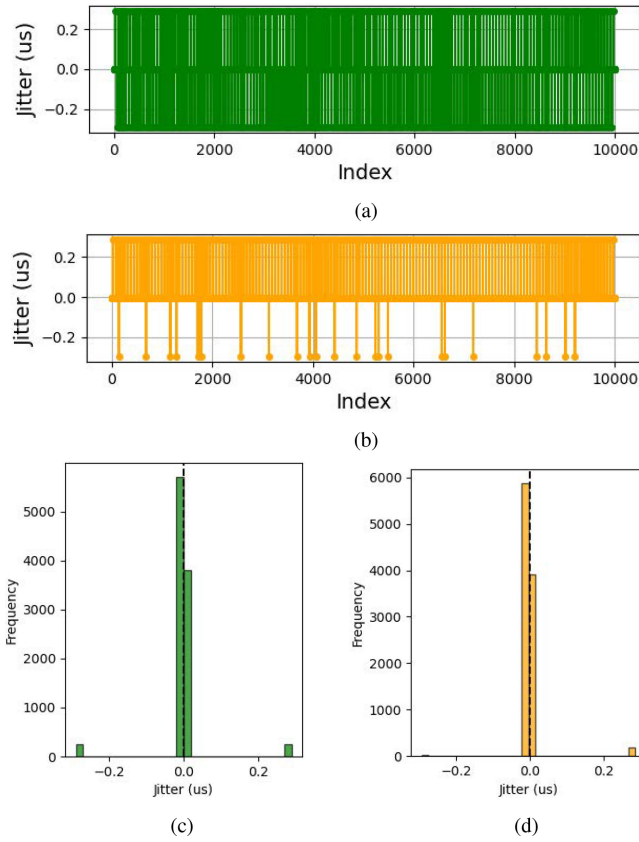


Fig. 9. Jitter FT and PT on DECT-2020. (a) Jitter at FT. (b) Jitter at PT. (c) Histogram of jitter at FT. (d) Histogram of jitter at PT.

interrupts, schedules tasks, and performs periodic system checks. Additionally, the inherent limitations of the RPi's hardware clock contribute to occasional timing delays.

Both Figs. 9 and 10 confirm that DECT has a significantly superior clock compared to the RPi's. The histograms indicate that DECT achieves jitter levels mostly around $0.1 \mu\text{s}$ (10^{-6} s), whereas RPi's exhibit jitter primarily around 0.1 ms (10^{-3} s), demonstrating that the DECT clock is *1000 times* more precise than that of the RPi's. These findings further highlight DECT's capability to support URLLC services for industrial IoT scenarios. Furthermore, the combination of DECT and Wi-Fi offers a solution that combines DECT's low-latency performance with Wi-Fi's high transmission speeds, providing enhanced wireless connectivity for IoT devices.

D. Testing 3: Bandwidth

To evaluate the proposed EWDC framework, the Wi-Fi performance was tested with and without WireGuard VPN enabled, using the RPi's embedded 802.11ac Wi-Fi on the 5.0 GHz band. First, ping latency was measured under both conditions. Next, the iPerf3¹⁰ was used to simulate data transfers by generating UDP traffic.

As shown in Fig. 11, the RPi system effectively detects the WireGuard VPN setup and teardown, as WireGuard operates as a Linux kernel-level interface. In this configuration, the WireGuard VPN is applied to the server-side IP address

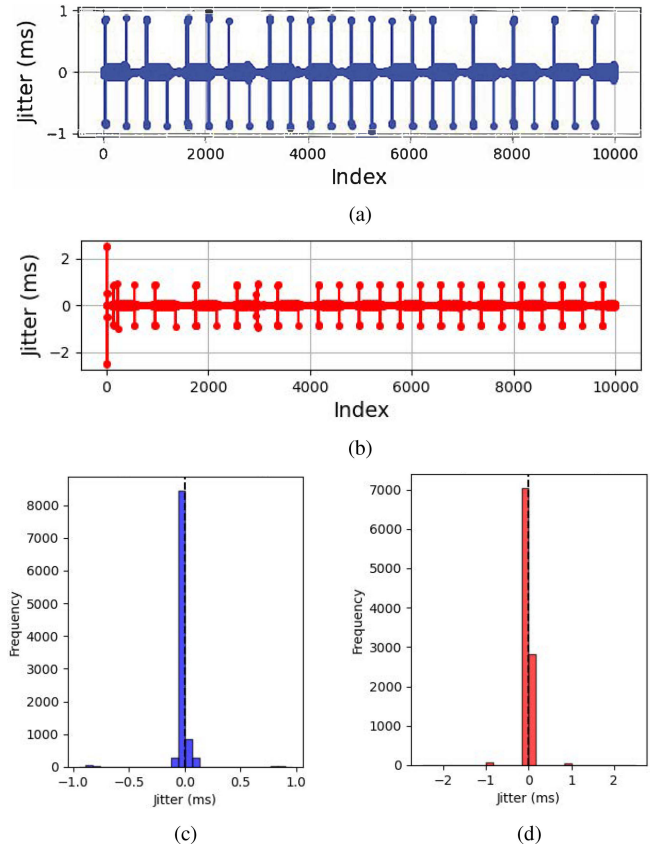


Fig. 10. Jitter at RPi-FT and RPi-PT. (a) Jitter at RPi_FT. (b) Jitter at RPi_PT. (c) Histogram of jitter at RPi-FT. (d) Histogram of jitter at RPi-PT.

10.10.0.11 and the client-side IP address 10.10.0.22. These IPs are configured using NAT and IP table rules to forward traffic to the WLAN interfaces utilizing Wi-Fi IPs 10.42.0.1 (server) and 10.42.0.197 (client). Fig. 11(e) demonstrates the negative impact with WireGuard VPN on ping latency and jitter. Specifically, with WireGuard VPN, the average ping time increases slightly by approximately 0.5 ms, from 7.07 to 7.52 ms, while jitter becomes more pronounced, rising from 0.51 to 1.33 ms. Interestingly, some individual ping latencies are lower with the VPN than without the VPN, likely due to Wi-Fi's contention-based transmission, which introduces randomness in packet transfers. However, the maximum ping latency exhibits a significant difference, increasing by around 3 ms with VPN compared to it without VPN. This behavior highlights the computational overhead introduced by WireGuard's encryption and decryption processes, particularly in high-demand or dynamic network environments.

Fig. 12 illustrates the iPerf 3 testing results, showcasing both transmission speed and transmission time for data increments of 10, 50, 100, 150, and 200 MB. Regarding transmission speed, enabling the VPN reduces the data transfer rate due to the additional computational overhead introduced by encryption. The transmission speed reaches approximately 60 Mb/s when the VPN is disabled, whereas it drops to around 50 Mb/s with the VPN enabled. The trend lines in the figure suggest that larger data sizes slightly impact transmission speed negatively, likely because encryption requires

¹⁰<https://iperf.fr/>

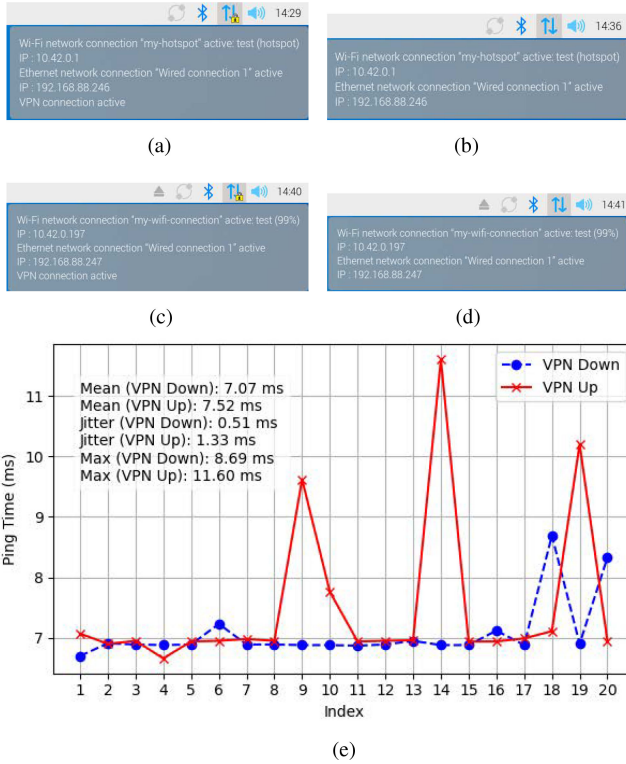


Fig. 11. Ping results with VPN down and VPN up. (a) Server with VPN. (b) Server without VPN. (c) Client with VPN. (d) Client without VPN. (e) Ping results.

additional computation, leading to system delays. In terms of transmission time, as expected, larger data sizes result in longer transmission durations. The VPN-enabled configuration shows a significantly higher transmission time compared to the VPN-disabled setup, attributed to the encryption processes consuming more time and computational resources. Although WireGuard introduces additional latency and reduces throughput, it provides robust data security over Wi-Fi.

E. Comparison Between EWDC and AWDL

To further evaluate EWDC, we compare it with AWDL in terms of latency, jitter, security, reliability, and energy efficiency.

1) *Latency and Jitter*: To evaluate the performance of the EWDC system, we compared latency and jitter measurements of DECT-2020 with publicly available BLE measurements¹¹ and Wi-Fi data reported in [32]. The BLE evaluation, using the Nordic nRF52832 SoC for low-latency MIDI applications, reported approximately 6 ms latency with around 1 ms jitter. For Wi-Fi (IEEE 802.11ac), the 99th-percentile latency indicated that 99% of packets experienced latencies below 8 ms in single-user scenarios, also with approximately 1 ms jitter. In contrast, DECT-2020 achieved significantly lower latency, ranging between 0.6 and 0.9 μ s, with jitter around 0.1 μ s. The higher latency and jitter in BLE result primarily from its use of connection intervals, during which devices periodically

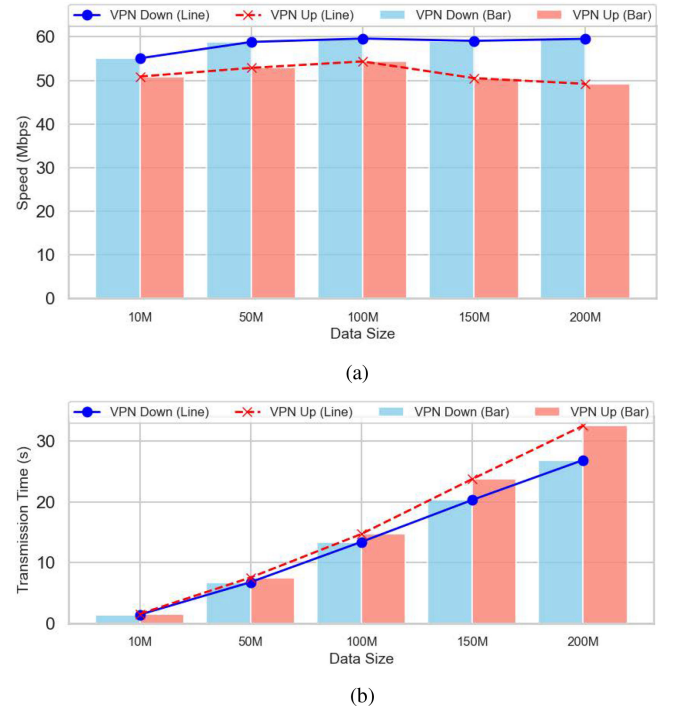


Fig. 12. iPerf3 transmission speed and time results. (a) Transmission speed. (b) Transmission time.

wake, exchange data, and return to sleep. Similarly, Wi-Fi employs the CSMA/CA mechanism, requiring collision avoidance before packet transmission, introducing additional latency and jitter. DECT-2020, however, utilizes deterministic transmissions and stabilized associations, significantly reducing latency and jitter compared to BLE and Wi-Fi. These results clearly demonstrate that DECT-2020 provides substantially higher timing precision, outperforming both BLE and Wi-Fi, whose performance remains in the millisecond range. Consequently, the proposed EWDC framework achieves notably lower latency and jitter for device association compared to AWDL.

2) *Security and Reliability*: In the AWDL authentication process, several sensitive data fields, including altDsID and encDsID that are linked to the user's Apple email addresses and phone numbers, are exchanged over the air along with validity durations and timestamps [2]. Although these data arrays are hashed using SHA-256 to represent validated contact information, the standard validity duration of 30 days raises concerns about privacy exposure [33]. Additionally, SHA-256 hashing has been shown to be insufficient for privacy protection, as hash values can be reversed using brute-force attacks or database lookups, thereby compromising user anonymity and confidentiality [34], [35]. Furthermore, since AWDL utilizes BLE for exchanging this information, it inherits BLE's inherent vulnerabilities, which could threaten data security and potentially lead to information leakage [6].

In contrast, EWDC employs DECT-2020 instead of BLE for device discovery and information exchange, significantly

¹¹<https://github.com/BLE-MIDI/NCS-MIDI>

enhancing security. Rather than exchanging user-related identifiers via hashes, EWDC transmits only temporary, single-use Wi-Fi credentials that change with each new connection setup, minimizing the risk of information leakage. For securing data transmissions, EWDC implements authenticated encryption using the ChaCha20-Poly1305 cipher alongside a WireGuard-based handshake. ChaCha20-Poly1305, a stream cipher combined with an authenticator, is specifically designed for high-speed, embedded IoT applications, providing efficient, lightweight, and robust encryption [36]. Unlike hash functions that provide only irreversible data integrity checks, stream ciphers encrypt data bit-by-bit in real-time, ensuring confidentiality and adaptability for resource-constrained IoT devices [37]. These design enhancements make EWDC a more secure and reliable alternative to AWDL for IoT data transmission.

3) *Energy Efficiency*: Energy efficiency is essential for extending device lifetimes in battery-powered IoT applications. In the EWDC framework, DECT-2020 replaces BLE for device discovery and authentication, while Wi-Fi continues to handle data transmission. Wi-Fi, though offering high throughput, consumes roughly 800 mW during transmission, 300–400 mW in idle mode, and about 20–40 mW in low-power sleep mode [38]. BLE is optimized for low power, drawing only a few milliamps during active states and approximately 1 μ A in sleep mode [39]. However, BLE has limitations regarding coverage and data rate. DECT-2020 addresses these limitations by providing a balanced energy profile, consuming around 45 mA in active (Rx/LBT) mode and only 8 μ A in sleep mode, along with flexible transmit power configurations [26], [27]. Compared to BLE, DECT-2020 exhibits higher energy consumption in active mode but significantly surpasses BLE in data transmission rates, achieving approximately 1.2 Gb/s as defined by ETSI DECT-2020, compared to BLE's maximum rate of 2 Mb/s as specified by Bluetooth SIG (see Table I). Moreover, DECT-2020 operates in a licensed frequency band (1.8 GHz), which is lower than BLE's and Wi-Fi's 2.4 and 5.0 GHz bands. Operating at lower frequencies reduces signal attenuation and path loss, enabling lower transmit power and fewer retransmissions, which improves energy efficiency [40]. Additionally, the use of a licensed band reduces interference, minimizes MAC-layer retransmissions, and further enhances the overall energy efficiency of the EWDC system.

Overall, replacing BLE with DECT-2020 in the proposed EWDC framework significantly improves latency and jitter performance, enhances security and reliability, and increases energy efficiency compared to AWDL. By utilizing DECT-2020, EWDC achieves deterministic low-latency communication, making it particularly suited for URLLC applications. Additionally, DECT-2020 operates in a licensed frequency band, substantially reducing interference and enhancing communication reliability compared to BLE-based AWDL. Moreover, EWDC adopts ChaCha20-Poly1305 encryption instead of SHA-256 hashing, ensuring stronger and more efficient data protection. These combined advantages position EWDC as a robust, secure, and energy-efficient solution for next-generation IoT communication systems.

VI. CONCLUSION AND FUTURE WORKS

This article proposed the EWDC framework, which integrates DECT-2020 NR with Wi-Fi to enable secure, efficient device discovery and high-throughput data transmission. Real-world implementation using Nordic nRF9161 hardware demonstrated that DECT-2020 achieves microsecond-level precision, while Wi-Fi provides higher data throughput. Additionally, WireGuard is employed to encrypt Wi-Fi data, enhancing security significantly despite the added latency and reduced throughput caused by encryption overhead. The observed ping delay highlights the potential to further optimize the EWDC system by combining DECT's deterministic transmission with Wi-Fi's high-data throughput at the MAC layer.

Future work could focus on optimizing the integration of DECT's deterministic transmission with Wi-Fi's high-speed capabilities at the MAC layer to minimize contention and jitter. Additionally, DECT's robust support for mMTC and mesh networking could further extend DECT-Wi-Fi mesh networks by acting as a reliable control plane that reduces contention, enhances bandwidth utilization, and improves overall transmission efficiency. Moreover, DECT Release 2, introduced in October 2024, introduces AES-128 encryption for ciphering and CMAC for integrity checking, which provides robust security measures that can further enhance the protection of large, decentralized IoT deployments [12]. Incorporating lightweight cryptographic techniques and robust key management systems will ensure secure communication with minimal computational overhead, especially in dynamic and rapidly evolving time-sensitive mesh networks.

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